

HARSHA VARDHAN DOPPALAPUDI

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Education

International Institute of Information Technology Bangalore

Jul 2024 – May 2028

Bachelor of Technology in Data Science and Artificial Intelligence

GPA: 3.34 / 4.0

Experience

NLP Research Intern (SRIP)

Jun 2026 – Present

IIIT Bangalore – Natural Language Processing Lab | Advisor: Prof. Tulika Saha

Bangalore, India

- Architected an autonomous LLM data-extraction pipeline with AI-as-a-Judge validation over the OpenAI API, converting **60,000+** words of unstructured anthropological text into **4,785** verified cultural rules of thumb across **9** Indian states, achieving **95%+** extraction precision.
- Spearheaded development of a cross-cultural evaluation benchmark spanning **29** Indian regional cultures to systematically measure Western bias in LLMs, targeting a top-tier peer-reviewed NLP venue.

Projects

SynthoGen AI: Privacy-Preserving Synthetic EHR | *PyTorch, Opacus, SDMetrics, Streamlit*

Jun 2026

[GitHub](#) | [Dashboard](#) | [IEEE DataPort Publication](#)

- Evaluated **4** generative architectures (TabSyn, TabDDPM, CTGAN, TVAE) across **12** model-dataset configurations on **258K+** patient records, delivering **94.6%** downstream ML utility (TSTR accuracy) with **zero** privacy breaches.
- Integrated DP-SGD differential privacy (Opacus) with formal auditing (DCR, K-Anonymity, Re-ID Risk) and shipped an interactive Streamlit dashboard; published synthetic datasets on IEEE DataPort (DOI: 10.21227/64c7-vj34).

Question Answering System | *PyTorch, Hugging Face Transformers, FastAPI, Docker*

Mar – Apr 2026

[GitHub](#) | [Live Demo](#)

- Pre-trained a **110M**-parameter Transformer encoder from scratch on Wikipedia and fine-tuned on **130K+** SQuAD v2 pairs, surpassing DistilBERT and T5-Small baselines with **58.2** F1 (Extractive) and **39.9** F1 (Generative).
- Slashed inference latency by **50%** through FP16 quantization and containerized the model as a Docker-based FastAPI REST API on Hugging Face Spaces, serving **100+** daily queries on free-tier CPU.

Traffic Violation Detection System | *YOLOv11, ONNX Runtime, FastAPI, Docker, GitHub Actions*

Apr – Jun 2026

[GitHub](#) | [Live Demo](#)

- Engineered a **3**-stage YOLOv11 cascade pipeline on **17,670** annotated images for real-time detection of helmetless riders, overloaded motorcycles, and license plates, achieving mAP50 scores of **0.936**, **0.829**, and **0.761**.
- Boosted inference throughput by **1.8×** via ONNX Runtime conversion and deployed a production-grade FastAPI service with end-to-end CI/CD (Docker + GitHub Actions) on Hugging Face Spaces.

Customer Lifetime Value Intelligence Engine | *Python, LightGBM, XGBoost, Optuna, Streamlit*

Apr – Jun 2025

[GitHub](#) | [Dashboard](#)

- Built a stacked LightGBM, XGBoost, and BG/NBD ensemble on **775K+** retail transactions, cutting CLV prediction MAE by **43.6%** through Optuna hyperparameter search and uncovering **424** high-value persuadable customers via uplift modeling.
- Shipped a production Streamlit dashboard featuring **85.6%**-coverage conformal prediction intervals, automated PSI-based drift alerts, and SHAP explainability, directly informing stakeholder retention strategies.

Technical Skills

Languages: Python, C++, SQL, Bash

ML Frameworks: PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, Optuna

Libraries & Tools: Hugging Face (Transformers, Datasets, Tokenizers), NumPy, Pandas, OpenCV, ONNX Runtime, SHAP, Matplotlib, Plotly, TensorBoard, Jupyter, OpenAI API

Deployment & APIs: FastAPI, REST APIs, Docker, Streamlit, GitHub Actions (CI/CD), Git, Linux

Relevant Coursework

- | | | | |
|--------------------|-------------------|---------------------------|----------------------------|
| • Machine Learning | • NLP | • Reinforcement Learning | • Probability & Statistics |
| • Deep Learning | • Computer Vision | • Data Structures & Algos | • Linear Algebra |

Achievements

- IEEE DataPort Publication:** Published privacy-preserving synthetic EHR datasets via latent diffusion (TabSyn); DOI: 10.21227/64c7-vj34.
- Kaggle – Currently Top 9% (Rank 217 / 2,400+):** Stellar Classification challenge; built a stacked ensemble of LightGBM, XGBoost, and CatBoost with balanced-accuracy optimization, anchor blending, and pseudo-label filtering.
- Datathon Finalist – Data Driven Policy Innovation** (IIIT Bangalore, Synergy '25): advanced to finals among **333** registered teams, developing a data-driven policy analysis solution.